Advanced Programming Practice · 21CSC203P

Unit 5 · Session 3 · SLO 1 — Solved Worksheet

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Unit â€“ 5 : Session â€“ 3 : SLO â€“ 1 | Implementing Automata with the Python automata Library

I. Questions & Answers

1. Library â€” the automata-lib library (pip install automata-lib), specifically automata.fa.dfa.DFA and automata.fa.nfa.NFA, is used to define DFAs/NFAs in Python.

2. Acceptance check â€” dfa.accepts_input("0101") returns True/False; dfa.read_input("0101") returns the final state.

3. Defining a transition â€” transitions are given as a nested dictionary: transitions={"q0": {"0": "q1", "1": "q0"}, "q1": {"0": "q1", "1": "q0"}} â€ i.e. Î´(state, symbol) = next state.

II. Example

• from automata.fa.dfa import DFA
• dfa = DFA(states={"q0", "q1"}, input_symbols={"0", "1"},
• transitions={"q0": {"0": "q1", "1": "q0"}, "q1": {"0": "q1", "1": "q0"}},
• initial_state="q0", final_states={"q1"})
• print(dfa.accepts_input("1010")) # True â€ string ends with 0